We have $\frac{dy}{dx} = \frac{y}{x} + \sqrt{1+\left(\frac{y}{x}\right)^2}$.

Let $y = vx$. Then, $v + x\frac{dv}{dx} = v + \sqrt{1+v^2}$, which becomes $x\frac{dv}{dx} = \sqrt{1+v^2}$.

By separating variables, we have $\frac{dv}{\sqrt{v^2+1}} = \frac{dx}{x}$.

Upon integration, we get $\log\left|v+\sqrt{v^2+1}\right| = \log|x| + \log|C|$ or $v+\sqrt{v^2+1} = xC$.

Now, we replace v by $\frac{y}{x}$, we get $\frac{y}{x} + \sqrt{\frac{y^2}{x^2}+1} = Cx$ (or) $y + \sqrt{x^2+y^2} = Cx^2$ gives the general solution of the given differential equation.

To determine the value of C, we use the condition that $y = 0$ when $x = 1$. So, we get $C = 1$.

Thus $y + \sqrt{x^2+y^2} = x^2$ is the particular solution of the given differential equation.

Example 10.19

Solve $(2x+3y)dx + (y-x)dy = 0$.

Solution

The given equation can be written as $\frac{dy}{dx} = \frac{2x+3y}{x-y}$.

This is a homogeneous equation.

Let $y = vx$. Then we have $v + x\frac{dv}{dx} = \frac{2+3v}{1-v}$.

Thus, $x\frac{dv}{dx} = \frac{2+2v+v^2}{1-v}$ or $\frac{1-v}{(1+v)^2+1}dv = \frac{dx}{x}$ or $-\frac{1}{2}\left[\frac{2v+2}{v^2+2v+2} - \frac{4}{(v+1)^2+1}\right]dv = \frac{dx}{x}$.

Integrating both sides, we get $-\frac{1}{2}\log\left|v^2+2v+2\right| + 2\tan^{-1}(v+1) = \log|x| + \log|C|$

or $\log\left|v^2+2v+2\right| - 4\tan^{-1}(v+1) = -2\log|x| - 2\log|C|$

or $\log\left|v^2+2v+2\right| + \log|x|^2 - 4\tan^{-1}(v+1) = -2\log|C|$

or $\log\left|(v^2+2v+2)x^2\right| - 4\tan^{-1}(v+1) = -2\log|C|$.

Now replacing v by $\frac{y}{x}$, we get, $\log\left|y^2+2xy+2x^2\right| - 4\tan^{-1}\left(\frac{x+y}{x}\right) = k$, where $k = -2\log|C|$ gives the required solution.

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